

THERMAL CONDUCTION IN SOLIDS



MARTIN

WELCOME!!

Electrical Engineering Materials (EEE2105)

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HEAT TRANSFER

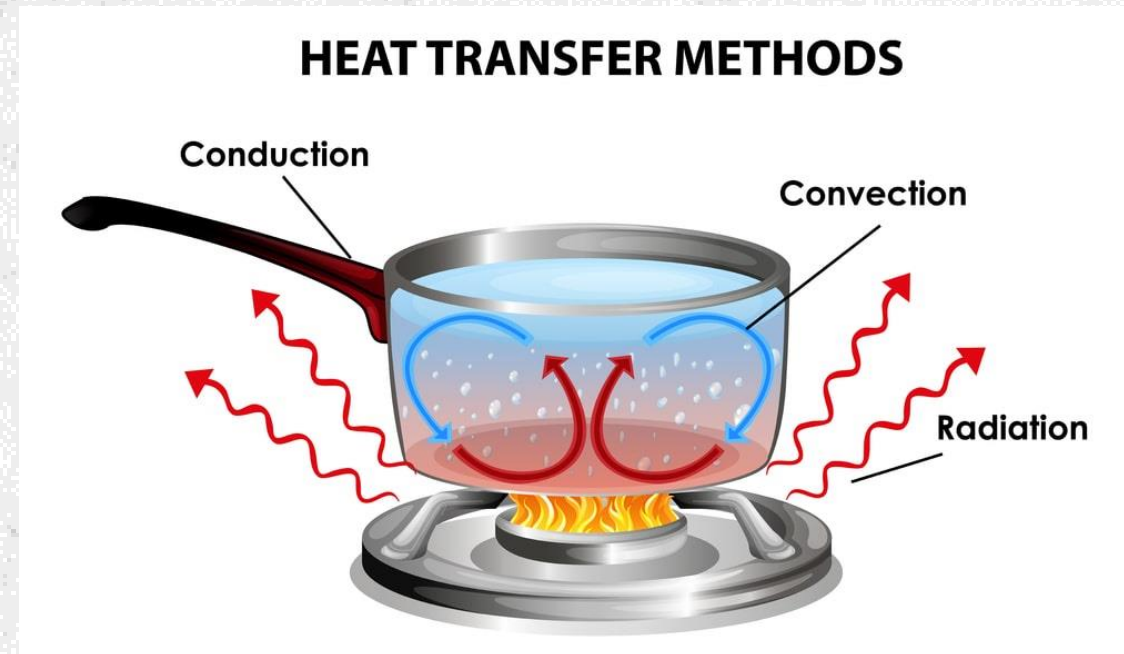
Heat transfer is the movement of thermal energy from one object or system to another due to a temperature difference.

Heat can be transferred through three mechanisms:

CONDUCTION occurs when there is direct contact between two objects at different temperatures.

CONVECTION involves the transfer of heat through the movement of fluids or gases.

RADIATION does not require a medium and can occur in a vacuum, as it involves the emission of electromagnetic waves by a hot object.

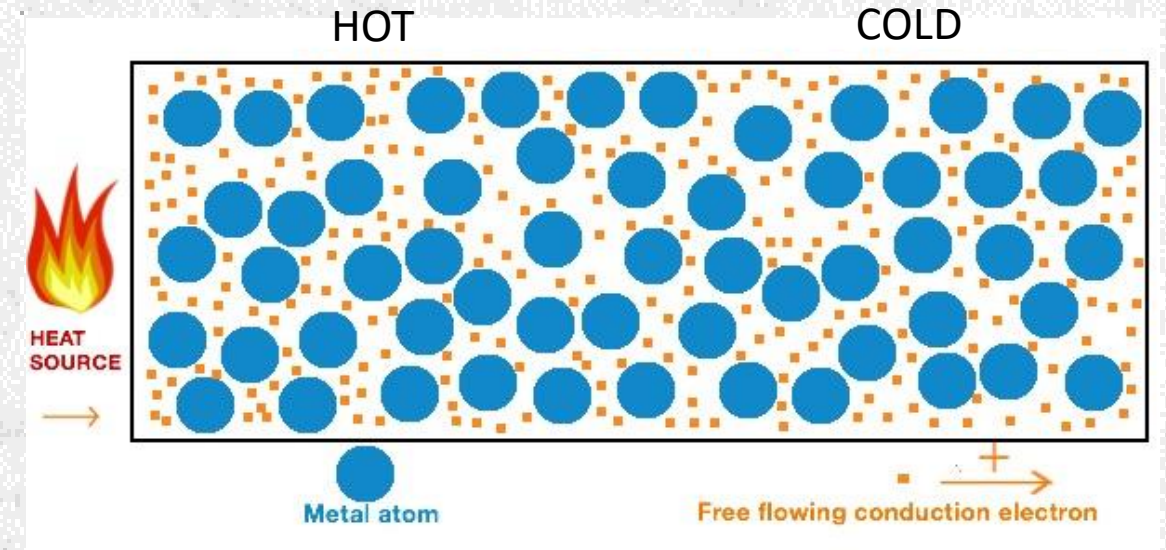
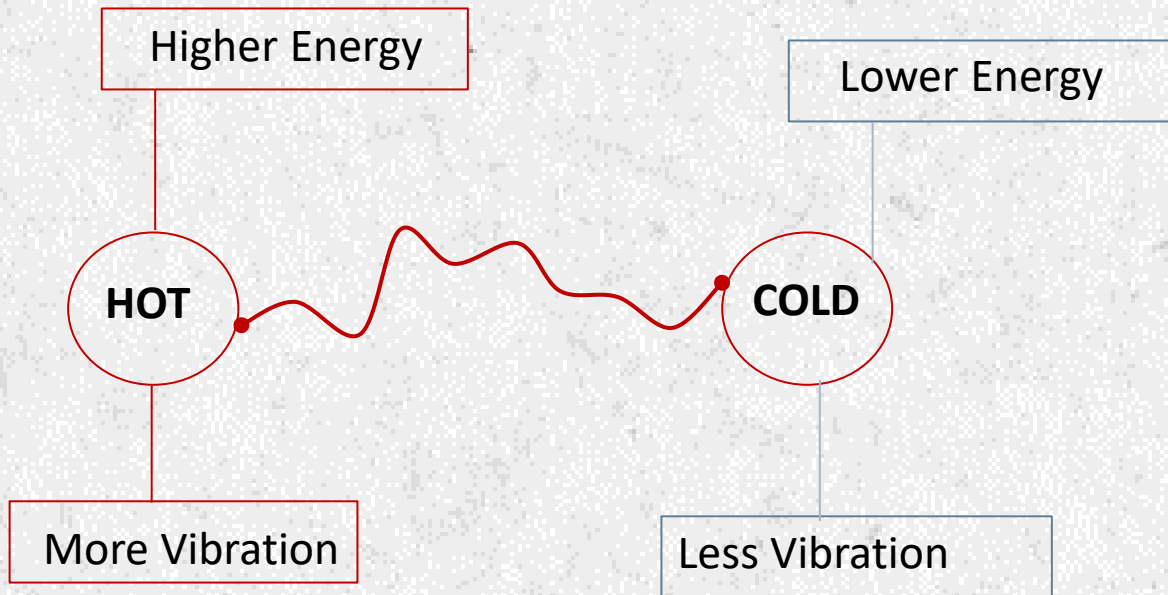


<https://pharmacygyan.com/what-is-heat-transfer/>

Fig 1: Heat Transfer Methods

THERMAL CONDUCTION

If there is a temperature gradient, heat will flow from higher temperature region. This is called thermal conduction.



<https://thermtest.com/physics-explanation-of-thermal-conductivity-of-metals>

Fig 2: Thermal Conduction

THERMAL CONDUCTION

Kinetic Energy :

$$E_{\text{avg}} = \frac{1}{4} M a^2 \omega^2$$

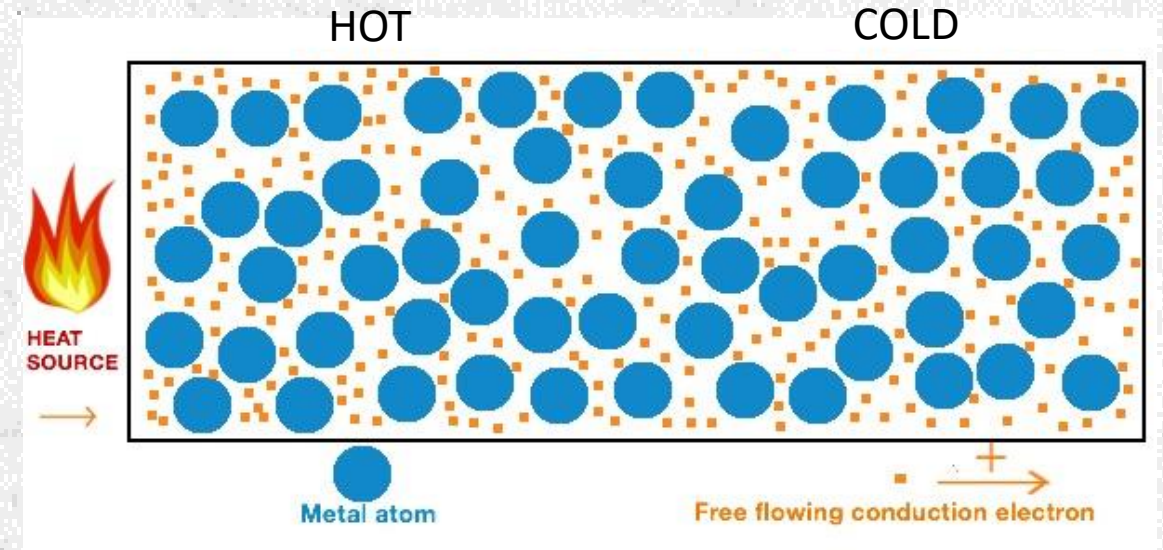
$$= \frac{1}{2} K T \quad [\text{Boltzmann constant } K]$$

T ↑

a ↑

E ↑

Increasing Temperature, causes the increase in amplitude of vibration of molecules and Energy of Molecules also increases.



<https://thermtest.com/physics-explanation-of-thermal-conductivity-of-metals>

Fig 2: Thermal Conduction

THERMAL CONDUCTION

Due to Collisions electron gain energy from energetic vibration.

By the virtue of increased scattering mechanism energetic electrons transfer extra energy to the colder region.

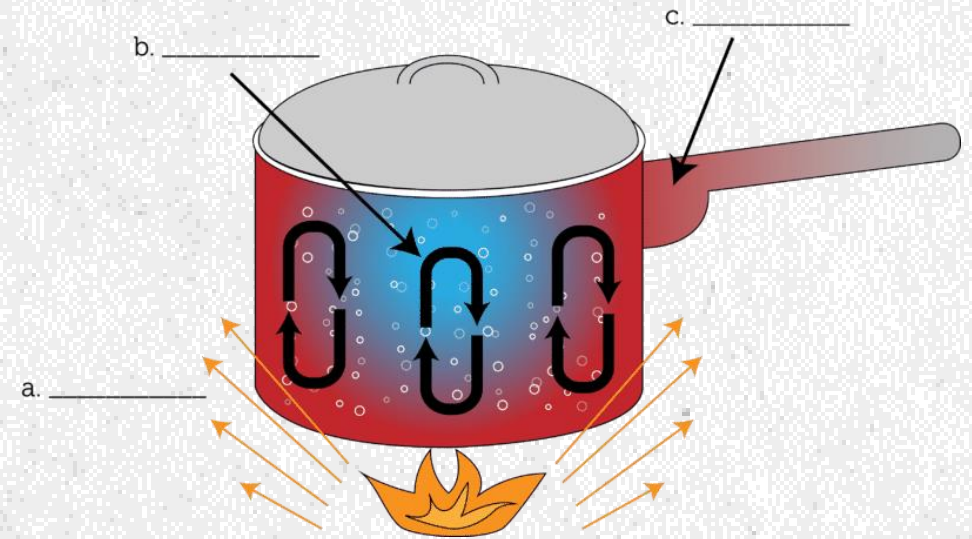


Fig 3: Thermal Conduction in different way

HEAT CONDUCTION IN METALS

Credit: www.doitpoms.ac.uk

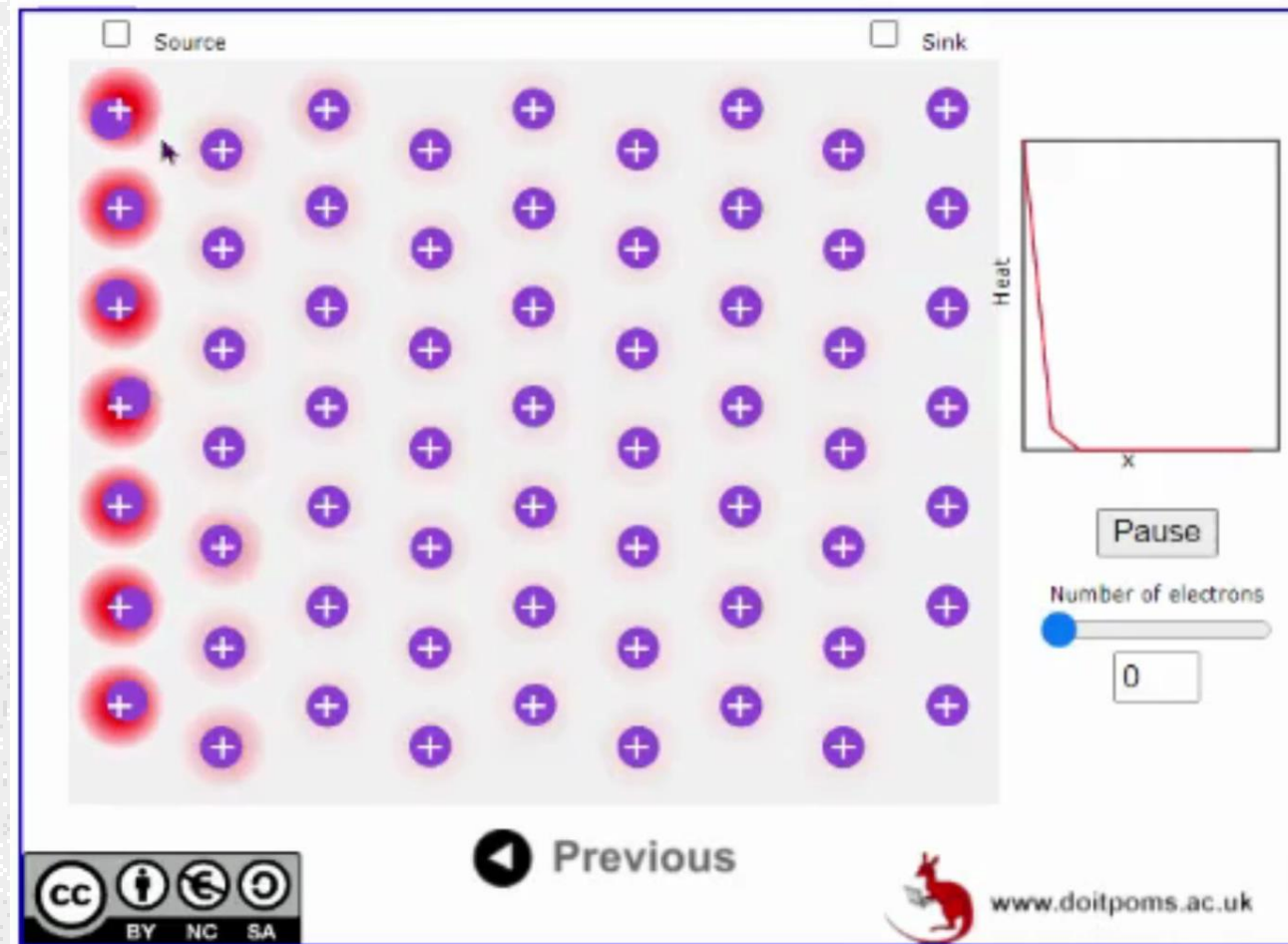


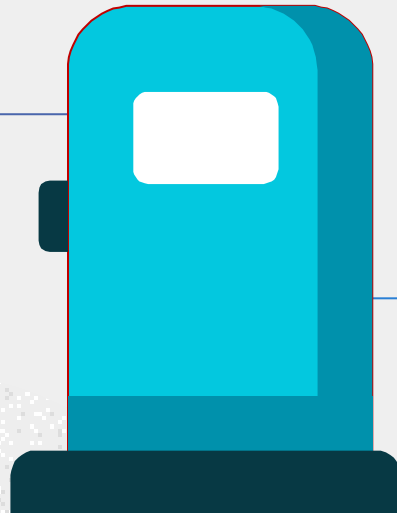
Fig 3: Heat conduction process

THERMAL CONDUCTIVITY

The ability of a material to transfer the heat energy is called the thermal conductivity.

Thermal Energy is transported in two ways:

Diffusion of
electrons



Transfer of
vibrating energy
(phonon)

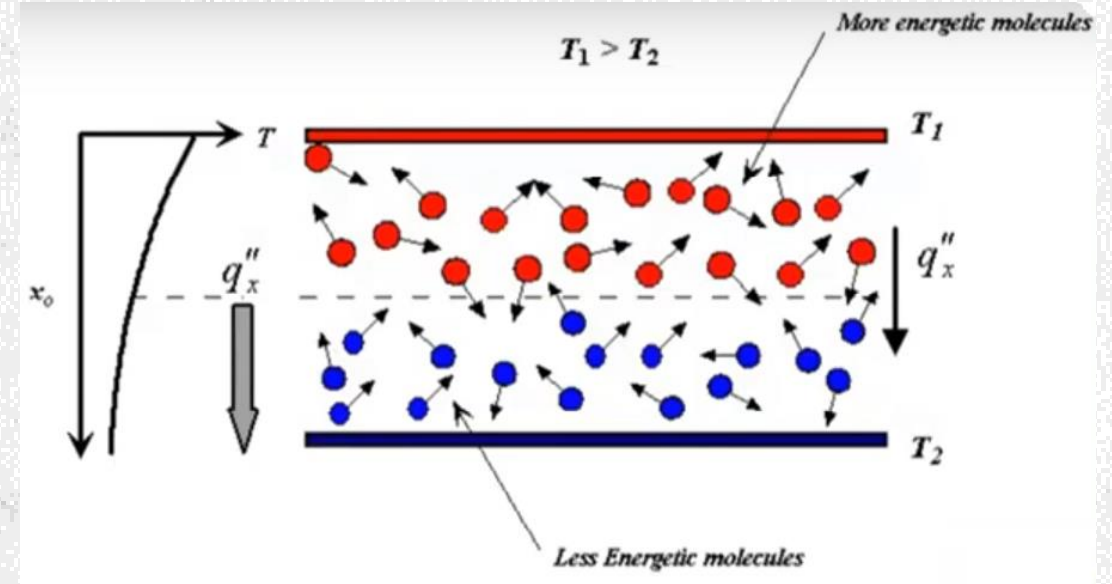


Fig 4: Thermal Conductivity

THERMAL CONDUCTIVITY

Diffusion of Electrons refers to the process by which electrons move from regions of higher concentration to regions of lower concentration, due to random thermal motion. This movement occurs in response to an electric potential gradient or a concentration gradient.

Transfer of vibrating energy or phonon refers to the quantized units of vibrational energy in solids, which propagate through a material in a manner similar to how photons propagate through a vacuum.

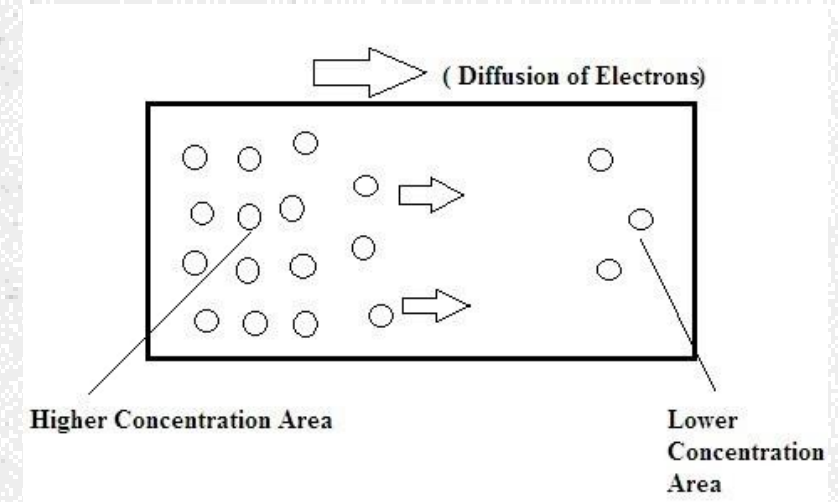


Fig 5: Diffusion of electrons

Photons are elementary particles of light that carry energy in the form of electromagnetic radiation, whereas **Phonons** are quanta of vibrational energy in lattice.

THERMAL CONDUCTIVITY

$$\kappa = \kappa_l + \kappa_e$$

κ = Thermal conductivity

κ_l = Lattice thermal conductivity

κ_e = Electrical thermal conductivity

κ is the material-dependent constant



THERMAL CONDUCTIVITY OF DIFFERENT MATERIALS

Typical thermal conductivity of various classes of materials at 25° C

Material	κ (W m ⁻¹ K ⁻¹)	Material	κ (W m ⁻¹ K ⁻¹)
Pure metal		Ceramics and glasses	
Nb	52	Glass-borosilicate	0.75
Fe	80	Silica-fused (SiO ₂)	1.5
Zn	113	S ₃ N ₄	20
W	178	Alumina (Al ₂ O ₃)	30
Al	250	Sapphire (Al ₂ O ₃)	37
Cu	390	Beryllium (BeO)	260
Ag	420	Diamond	~1000
Metal alloys		Polymers	
Stainless steel	12–16	Polypropylene	0.12
55% Cu–45% Ni	19.5	PVC	0.17
70% Ni–30% Cu	25	Polycarbonate	0.22
1080 steel	50	Nylon 6,6	0.24
Bronze (95% Cu–5% Sn)	80	Teflon	0.25
Brass (63% Cu–37% Zn)	125	Polyethylene, low density	0.3
Dural (95% Al–4% Cu–1% Mg)	147	Polyethylene, high density	0.5

Fig 6: Thermal Conductivity of different materials

FOURIER'S LAW OF THERMAL CONDUCTION

The rate of heat flow, $Q' = \frac{dQ}{dt}$ through a thin section of thickness δx is proportional to the temperature gradient $\frac{\delta T}{\delta x}$ and the cross section area A .

So,

$$Q' \propto \frac{\delta T}{\delta x} \dots\dots\dots (i)$$

$$Q' \propto A \dots\dots\dots (ii)$$

From (i) and (ii) $\Rightarrow$

$$Q' \propto A \frac{\delta T}{\delta x}$$

$$Q' = - \kappa A \frac{\delta T}{\delta x}$$

Where, κ is the thermal conductivity (W/m-k)

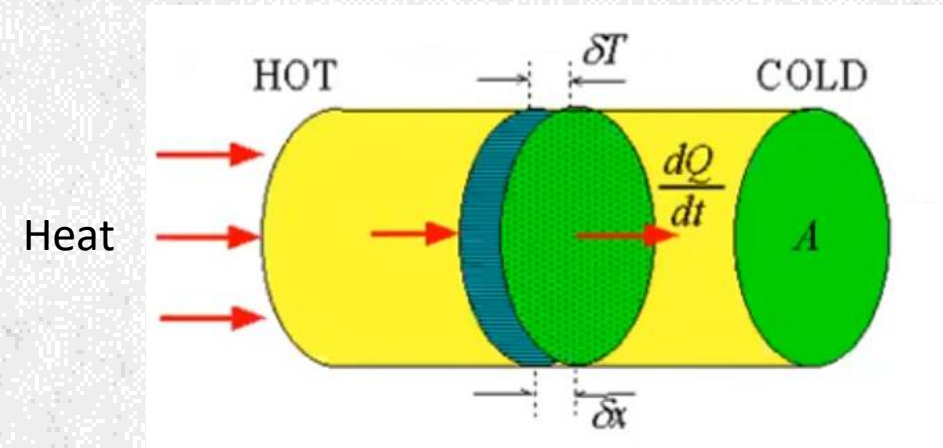


Fig 7: Heat flow in a metal rod heated at one end.

FOURIER'S LAW OF THERMAL CONDUCTION

💡 Rate of Heat Flow: $Q' = - \kappa A \frac{\delta T}{\delta x}$ (J/s)

where, $\frac{\delta T}{\delta x}$ is the temperature gradient or thermal gradient of the medium.

Similarly,

💡 Rate of Charge Flow: $I' = - \sigma A \frac{\delta V}{\delta x}$

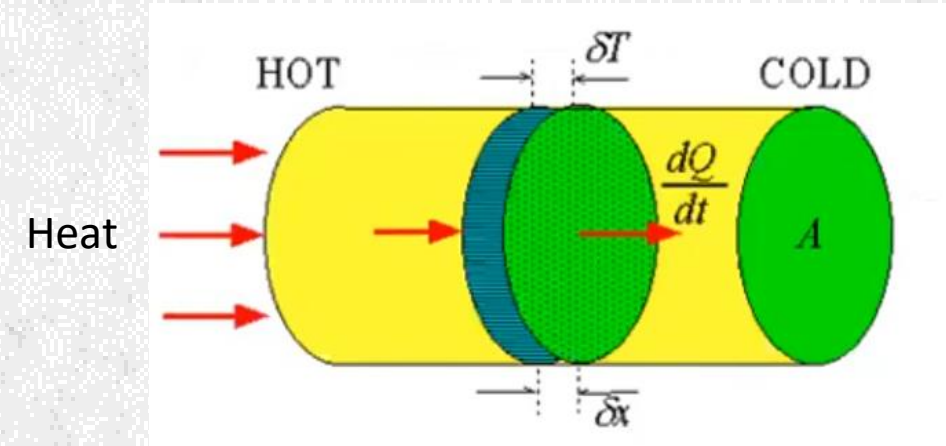


Fig 7: Heat flow in a metal rod heated at one end.

The negative sign(-) indicates that heat flows from regions of higher temperature to regions of lower temperature.

APPLICATION OF THERMAL CONDUCTION

INSULATION: Materials with low thermal conductivity are ideal for insulation purposes as they can prevent the transfer of heat from one area to another. For instance, materials like fiberglass, rock wool, and cellulose are widely used as insulations in buildings and industries.



Fig 8: Insulation

HEAT SINK: Materials with high thermal conductivity are used as heat sinks to absorb and dissipate heat from electronic devices such as CPUs, GPUs, and power amplifiers. Copper and aluminum are popular materials for heat sink applications due to their excellent thermal conductivity.



Fig 9: Heat Sink

AUTOMOTIVE INDUSTRY: Thermal conductivity is important in the automotive industry where materials must withstand high temperatures and have good heat dissipation properties. Aluminum and magnesium alloys, which have high thermal conductivity, are often used in engine blocks and cylinder heads.



Fig 10: Automotive Industry

APPLICATION OF THERMAL CONDUCTION

SOLAR PANELS: Solar panels require efficient absorption and transfer of solar energy into electrical energy. Materials with high thermal conductivity like silicon, copper, and aluminum are used as conductor layers in solar cells.

COOKWARE: Cookware manufacturers use materials with high thermal conductivity, such as copper and aluminum, to make pots and pans. These materials ensure that heat is evenly distributed across the cooking surface, leading to better cooking results.

In summary, the thermal conductivity in solids is critical in several fields, including building construction, electronics, automotive, energy, and manufacturing, etc. We can design the material considering there value of thermal conduction as well as thermal conductivity.

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- Assisting Youtube channel FN_AUST
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LET'S CLEAR ANY DOUBTS





THANK YOU
